```
import numpy as np
import pandas as pd
print("Numpy Version:",np.__version__)
print("Pandas Version:",pd.__version__)
```

```
Numpy Version: 2.0.2
Pandas Version: 2.2.3
```

```
marks=np.array([78,89,90,99,98,97])
print("marks:",marks)
print("Type of marks:",type(marks))
```

```
marks: [78 89 90 99 98 97]
Type of marks: <class 'numpy.ndarray'>
```

```
print("Dimension of marks:",marks.ndim)
print("Shape of marks:",marks.shape)
print("Size of marks:",marks.size)
```

```
Dimension of marks: 1
Shape of marks: (6,)
Size of marks: 6
```

```
marks=np.array([78,89,90,99,98,97])
print(" Orignal Marks:",marks)
print("Add 5:",marks+5)
print("Multifly by 2:",marks*2)
print("Divdded by 2:",marks/2)
print("Subtracted by 2:",marks-2)
print("Squre:",marks**2)
```

```
 Orignal Marks: [78 89 90 99 98 97]
Add 5: [ 83  94  95 104 103 102]
Multifly by 2: [156 178 180 198 196 194]
Divdded by 2: [39.  44.5 45.  49.5 49.  48.5]
Subtracted by 2: [76 87 88 97 96 95]
Squre: [6084 7921 8100 9801 9604 9409]
```

```
print("Sum:",marks.sum())
print("Min:",marks.min())
print("Max:",marks.max())
print("Mean:",marks.mean())
```

```
Sum: 551
Min: 78
Max: 99
Mean: 91.83333333333333
```

```
print("First Marks:",marks[0])
print("Second Marks:",marks[1])
print("Third Marks:",marks[2])
```

```
print("Last Marks:",marks[-1])
print("Second Last Marks:",marks[-2])
print("Third Last Marks:",marks[-3])
```

```
First Marks: 78
Second Marks: 89
Third Marks: 90
Last Marks: 97
Second Last Marks: 98
Third Last Marks: 99
```

```
marks=np.array([78,89,90,99,98,97])
print("Index 1 to 3:",marks[1:4])
print("Index 2 to end:",marks[2:])
print("Index 0 to 3:",marks[:4])
print("Every second element:",marks[::2])
```

```
Index 1 to 3: [89 90 99]
Index 2 to end: [90 99 98 97]
Index 0 to 3: [78 89 90 99]
Every second element: [78 90 98]
```

```
marks=np.array([
[1,2,3],
 [4,5,6],
  [7,8,9]])
print("Shape of the array:",marks.shape)
print("Mean of all values:",marks.mean())
print("Mean of each column:",marks.mean(axis=0))
print("Mean of each row:",marks.mean(axis=1))
```

```
Shape of the array: (3, 3)
Mean of all values: 5.0
Mean of each column: [4. 5. 6.]
Mean of each row: [2. 5. 8.]
```

```
import pandas as pd
data={
    "name":["sachin","sameer","rahul"],
    "Branch":["CSE","AIML","CSE"],
    "Marks":[98,99,97]
}
df=pd.DataFrame(data)
print(df)
print("Dimension of the:",df.shape)
print("\nColumns:")
print(df.columns)
```

```
     name Branch  Marks
0  sachin    CSE     98
1  sameer   AIML     99
2   rahul    CSE     97
Dimension of the: (3, 3)
```

```
Columns:
Index(['name', 'Branch', 'Marks'], dtype='object')
```

```
df.head()
```

	name	Branch	Marks
0	sachin	CSE	98
1	sameer	AIML	99
2	rahul	CSE	97

```
df.tail()
```

	name	Branch	Marks
0	sachin	CSE	98
1	sameer	AIML	99
2	rahul	CSE	97

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   name    3 non-null      object
 1   Branch  3 non-null      object
 2   Marks   3 non-null      int64 
dtypes: int64(1), object(2)
memory usage: 204.0+ bytes
```

```
df.describe()
```

	Marks
count	3.0
mean	98.0
std	1.0
min	97.0
25%	97.5
50%	98.0
75%	98.5
max	99.0

```
print(df["Marks"])
print(df[["Branch","Marks"]])
```

```
0    98
1    99
2    97
Name: Marks, dtype: int64
  Branch  Marks
0    CSE     98
1   AIML     99
2    CSE     97
```

```
print("First Row:")
print(df.iloc[0])
print("\nSecond row:")
print(df.iloc[1])
print("\nFrst three rows:")
print(df.iloc[0:3])
print("First two rows and two  column:")
print(df.iloc[0:2,0:2])
display(df.iloc[0:3,0:2])
```

```
First Row:
name      sachin
Branch       CSE
Marks         98
Name: 0, dtype: object

Second row:
name      sameer
Branch      AIML
Marks         99
Name: 1, dtype: object
```

```
high_scores=df[df["Marks"]>80]
print(high_scores)
```

```
  name Branch  Marks
0  sachin    CSE     98
1  sameer   AIML     99
2   rahul    CSE     97
First two rows and two column:
    name Branch
0  sachin    CSE
1  sameer   AIML
2   rahul    CSE
0  sachin    CSE
1  sameer   AIML
```

```
print("Students from CSE:")
display(df[df["Branch"]=="CSE"])
```

```
Students from CSE:
```

	name	Branch	Marks
0	sachin	CSE	98
2	rahul	CSE	97